

The Impact of Technology on Global and Local Economies

Global Economic Shifts: ECON 220

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Introduction

- ▶ Technology: the application of scientific knowledge for practical purposes, especially in industry.
- ▶ It is a key driver of economic growth, productivity, and societal change.
- ▶ Focus:
 - ▶ Global impacts of technology.
 - ▶ Local impacts on Northern British Columbia (NBC).

Defining Technological Change

- ▶ **Technological Change:** The process of invention, innovation, and diffusion of technology.
- ▶ Types:
 - ▶ **Incremental innovations:** Small improvements (e.g., better smartphone cameras).
 - ▶ **Radical innovations:** Disruptive technologies (e.g., AI, blockchain).

Historical Perspective

- ▶ Industrial Revolutions:
 - ▶ 1st: Steam power and mechanization.
 - ▶ 2nd: Electricity and mass production.
 - ▶ 3rd: Computers and automation.
 - ▶ 4th: AI, IoT, and advanced robotics.
- ▶ Each revolution disrupted labor markets, trade, and production.

Global Economic Impact Overview

- ▶ Technology impacts global economies through:
 - ▶ Productivity growth.
 - ▶ Trade facilitation.
 - ▶ Labor market transformations.
- ▶ Creates opportunities and challenges.

Productivity Growth: Explanation

- ▶ **Definition:** Productivity is the output per unit of input.
- ▶ Technology increases productivity by:
 - ▶ Automating tasks.
 - ▶ Enhancing accuracy and speed.

Example: Productivity Growth

- ▶ Example: Agricultural Technology
 - ▶ Tractors, GPS systems, and automated harvesters improve efficiency.
 - ▶ Result: Reduced labor costs and increased crop yields.
- ▶ Global Impact: More food supply and lower prices.

Trade Dynamics: Digital Platforms

- ▶ Platforms like Amazon, Alibaba, and Shopify reduce entry barriers.
- ▶ Example:
 - ▶ A small artisan in Kenya sells handmade products globally via Etsy.
- ▶ Result: Greater market access and income for small producers.

Trade Dynamics: Blockchain Technology

- ▶ Blockchain enhances supply chain transparency and reduces fraud.
- ▶ Example:
 - ▶ Walmart uses blockchain to track food origins, improving safety.

Labor Market Transformations

- ▶ Automation replaces routine jobs.
- ▶ Demand grows for tech-related roles (e.g., data analysts, software developers).
- ▶ Skills gap:
 - ▶ Many workers lack training for high-skill jobs.

Labor Market Example: AI in Finance

- ▶ AI automates tasks like fraud detection and credit risk assessment.
- ▶ Result: Increased efficiency but fewer entry-level analyst roles.
- ▶ New jobs: AI model developers and cybersecurity experts.

Economic Inequality

- ▶ Technology often benefits high-skill workers more than low-skill workers.
- ▶ Example:
 - ▶ Wealth concentration in tech hubs like Silicon Valley.

Local Impacts on NBC: Overview

- ▶ NBC's economy is heavily resource-driven.
- ▶ Technology impacts industries like forestry, mining, and energy.

Forestry: Technology Adoption

- ▶ GIS and drones improve sustainable logging.
- ▶ Example:
 - ▶ Canfor monitors forest health with satellite imagery.

Mining: Advanced Technologies

- ▶ Autonomous vehicles reduce operational risks.
- ▶ Example:
 - ▶ Mount Milligan mine uses advanced drilling technology.

Renewable Energy in NBC

- ▶ Hydroelectric projects like Site C integrate advanced storage.
- ▶ Clean tech startups innovate energy-efficient solutions.

Digital Connectivity: Challenges in NBC

- ▶ Rural and remote communities in NBC lack high-speed internet.
- ▶ Impacts:
 - ▶ Limited access to online education and healthcare.
 - ▶ Businesses face challenges in participating in e-commerce.

Connected Coast Project

- ▶ Aims to bring high-speed broadband to remote coastal communities.
- ▶ Benefits:
 - ▶ Enables remote work opportunities.
 - ▶ Promotes digital entrepreneurship.
- ▶ Example:
 - ▶ Prince Rupert businesses can access global markets.

Indigenous Participation in Technology

- ▶ Indigenous communities increasingly engage in tech-driven initiatives.
- ▶ Examples:
 - ▶ Digital storytelling platforms to preserve cultural heritage.
 - ▶ Collaboration with clean energy companies on renewable projects.
- ▶ Challenges:
 - ▶ Need for digital skills training.
 - ▶ Ensuring equitable access to technology.

Clean Tech and Green Economy in NBC

- ▶ Clean tech is an emerging sector in NBC.
- ▶ Example:
 - ▶ Hydrogen energy initiatives in Prince George.
- ▶ Benefits:
 - ▶ Diversifies the economy.
 - ▶ Aligns with global climate goals.

Tourism and Technology in NBC

- ▶ Technology enhances tourism in NBC:
 - ▶ Virtual reality tours for promoting attractions.
 - ▶ Online booking systems for remote lodges and activities.
- ▶ Example:
 - ▶ Virtual tours of the Great Bear Rainforest.

Challenges in Implementing Technology

- ▶ Key challenges in NBC:
 - ▶ High costs of infrastructure in remote areas.
 - ▶ Resistance to change in traditional industries.
 - ▶ Skills gap in the local workforce.

Skills Development for NBC Workforce

- ▶ Importance of upskilling the workforce:
 - ▶ Training programs for digital literacy.
 - ▶ Partnerships between industries and educational institutions.
- ▶ Example:
 - ▶ College of New Caledonia offers tech-focused courses.

Case Study: AI in Manufacturing (Global)

- ▶ AI-powered robotics improve efficiency and precision in manufacturing.
- ▶ Example:
 - ▶ BMW's smart factories use AI to streamline production.

Case Study: Renewable Energy in Iceland

- ▶ Iceland uses geothermal and hydropower for nearly 100% of electricity.
- ▶ Lessons for NBC:
 - ▶ Leverage abundant natural resources for clean energy leadership.

Case Study: E-Commerce in Southeast Asia

- ▶ E-commerce platforms like Lazada and Tokopedia drive regional growth.
- ▶ Opportunity for NBC:
 - ▶ Develop local e-commerce ecosystems for small businesses.

Case Study: Technology in Agriculture (India)

- ▶ Farmers use apps for market prices, weather updates, and crop health.
- ▶ Example:
 - ▶ The "eNAM" platform connects farmers to markets digitally.
- ▶ Potential for NBC:
 - ▶ Develop similar tools for forestry and agriculture.

Case Study: Tech Hubs in Africa

- ▶ Growth of innovation hubs (e.g., Nairobi's Silicon Savannah).
- ▶ Lessons for NBC:
 - ▶ Foster collaboration between government, academia, and private sector.

Future Trends in Technology

- ▶ Emerging trends:
 - ▶ Artificial intelligence and machine learning.
 - ▶ Blockchain for supply chains.
 - ▶ Renewable energy breakthroughs.

Global Collaboration

- ▶ Importance of global partnerships in technology.
- ▶ Examples:
 - ▶ AI regulation frameworks.
 - ▶ Sharing clean energy innovations.

Policy Recommendations for NBC

- ▶ Invest in digital infrastructure.
- ▶ Create incentives for tech startups.
- ▶ Support clean energy and sustainable industries.

Conclusion

- ▶ Technology reshapes economies globally and locally.
- ▶ NBC has opportunities in clean energy, connectivity, and resource innovation.
- ▶ Strategic policies are needed for inclusive growth.

Discussion Questions (1/3)

1. How can NBC better balance resource extraction with environmental sustainability?

Discussion Questions (2/3)

2. What specific skills are most critical for NBC's workforce to develop?

Discussion Questions (3/3)

3. What lessons can NBC learn from global success stories in technology?